

Closed Cones Are Exactly the Conic Fitzpatrick Fixed Points

Run `fa_banach_001` — lane 5

9 August 2026

Status: partial result, likely valid.

1 Source problem and scope

García Ramos, Martínez-Legaz, and Simons ask for a characterization of the sets A satisfying $G_{\Phi_A} = A$ in a symmetrically self-dual (SSD) space, explicitly presenting this as a generalized version of Fitzpatrick’s open monotone-operator problem [1, 2]. Their exact formulation appears in Section 3.1 on PDF page 5:

3.1 On a problem posed by Fitzpatrick

Let B be an SSD space and $f : B \rightarrow \mathbb{R} \cup \{+\infty\}$ be a proper convex function. The generalized Fenchel-Young inequality establishes that

$$f(a) + f^{\circledast}(b) \geq [a, b], \quad \forall a, b \in B. \quad (5)$$

We define the *q-subdifferential* of f at $a \in B$ by

$$\partial_q f(a) := \{b \in B : f(a) + f^{\circledast}(b) = [a, b]\}$$

and the set

$$G_f := \{b \in B : b \in \partial_q f(b)\}.$$

In this Subsection we are interested in identifying sets $A \subset B$ with the property that $G_{\Phi_A} = A$. The problem of characterizing such sets is an abstract version of an open problem on monotone operators posed by Fitzpatrick [3, Problem 5.2].

This packet solves the problem completely for the subclass of cones in the positive-definite Hilbert SSD model. It does not solve the general problem in an indefinite SSD space or the classical monotone-operator problem on $X \times X^*$.

2 Definitions

Let H be a real Hilbert space, paired with itself by its inner product, and put

$$q(x) = \frac{1}{2}\|x\|^2.$$

For a nonempty set $A \subset H$, write $d_A(x) = \inf_{a \in A} \|x - a\|$ and define

$$\Phi_A(x) = q(x) - \inf_{a \in A} q(x - a) = \sup_{a \in A} \{\langle x, a \rangle - q(a)\}.$$

For a proper convex function f , its intrinsic conjugate is the ordinary Fenchel conjugate

$$f^*(y) = \sup_{z \in H} \{\langle z, y \rangle - f(z)\}.$$

The q -subdifferential and its diagonal fixed-point set are

$$\partial_q f(x) = \{y : f(x) + f^*(y) = \langle x, y \rangle\}, \quad G_f = \{x : x \in \partial_q f(x)\}.$$

A cone means a nonempty set A such that $tA \subset A$ for every $t \geq 0$; no convexity is assumed.

3 Idea of the proof

The fixed-point condition has an exact metric form involving the squared distance to A . A cone makes this distance homogeneous along every ray. At a point $x \notin A$, moving a little farther out on the same ray changes d_A^2 to first order, while the penalty $\|b - x\|^2$ changes only to second order. Consequently the metric equality required for $x \in G_{\Phi_A}$ fails. Closedness is exactly what makes $d_A(x) > 0$ outside A . Conversely, every point in $\text{cl } A$ automatically satisfies the fixed-point equality, so a conic fixed point cannot be nonclosed.

4 Metric fixed-point criterion

Lemma 1. *For every nonempty $A \subset H$ and $x \in H$,*

$$x \in G_{\Phi_A} \iff \sup_{b \in H} \{d_A(b)^2 - \|b - x\|^2\} = d_A(x)^2.$$

Proof. Completing the square gives

$$\Phi_A(x) = \frac{1}{2} \|x\|^2 - \frac{1}{2} d_A(x)^2.$$

Therefore

$$\begin{aligned} \Phi_A^*(x) &= \sup_{b \in H} \left\{ \langle b, x \rangle - \frac{1}{2} \|b\|^2 + \frac{1}{2} d_A(b)^2 \right\} \\ &= \frac{1}{2} \|x\|^2 + \frac{1}{2} \sup_{b \in H} \{d_A(b)^2 - \|b - x\|^2\}. \end{aligned}$$

By definition, $x \in G_{\Phi_A}$ exactly when $\Phi_A(x) + \Phi_A^*(x) = \|x\|^2$. Substitution yields the stated equality. Notice that the supremum is always at least $d_A(x)^2$, by taking $b = x$. $\square$

5 Closed-cone classification

Theorem 1. *Let A be a nonempty cone in a real Hilbert space H . Then*

$$A = G_{\Phi_A} \iff A \text{ is norm closed.}$$

In particular, every nonempty closed cone, whether convex or not, is a generalized Fitzpatrick fixed point.

Proof. First suppose $A = G_{\Phi_A}$. Let $x \in \text{cl } A$. Then $d_A(x) = 0$, and for every $b \in H$,

$$d_A(b) = d_{\text{cl } A}(b) \leq \|b - x\|.$$

Thus the supremum in Lemma 1 is at most zero; taking $b = x$ shows that it equals zero. Hence $x \in G_{\Phi_A} = A$. We have $\text{cl } A \subset A$, so A is closed. This direction in fact does not use the cone assumption.

Conversely, suppose A is closed. Since A is a nonempty cone, $0 \in A$. If $x \in A$, then $d_A(x) = 0$ and $d_A(b) \leq \|b - x\|$ for every b , so Lemma 1 gives $x \in G_{\Phi_A}$.

Now let $x \notin A$ and set $\delta = d_A(x) > 0$. For every $\lambda > 0$, the cone identity $\lambda A = A$ implies

$$d_A(\lambda x) = \lambda d_A(x) = \lambda \delta.$$

Choose $\lambda > 1$ sufficiently close to 1 that

$$(\lambda + 1)\delta^2 > (\lambda - 1)\|x\|^2;$$

such a choice exists because the left side tends to $2\delta^2 > 0$ and the right side tends to zero. Evaluating the supremum in Lemma 1 at $b = \lambda x$ gives

$$\begin{aligned} \sup_{b \in H} \{d_A(b)^2 - \|b - x\|^2\} &\geq \lambda^2 \delta^2 - (\lambda - 1)^2 \|x\|^2 \\ &= \delta^2 + (\lambda - 1)((\lambda + 1)\delta^2 - (\lambda - 1)\|x\|^2) \\ &> \delta^2 = d_A(x)^2. \end{aligned}$$

The criterion therefore gives $x \notin G_{\Phi_A}$. We have proved $G_{\Phi_A} = A$. □

6 Consequences and relation to the source

The source proves that every closed convex set is a fixed point, and then checks directly that

$$\{(x_1, x_2) \in \mathbb{R}^2 : x_1 x_2 = 0\}$$

is a nonconvex fixed point. That set is a closed cone, so Theorem 1 both recovers the example and extends it to arbitrary closed cones in arbitrary real Hilbert spaces. It also supplies the converse within the conic class: no nonclosed cone can be a fixed point.

The theorem does not characterize arbitrary closed nonconic sets, and the general Fitzpatrick problem remains open outside the stated subclass.

7 Verification and novelty check

The proof uses only the displayed definitions, completion of the square, distance homogeneity for cones, and the elementary fact that a point outside a closed set has positive distance. Edge cases $A = \{0\}$ and $A = H$ are included. No finite-dimensionality, convexity, or projection theorem is used.

A bounded search through 9 August 2026 checked the run’s registry, solution, attempt, and proof-gap indexes; the exact source title and citations; and combinations of “closed cone”, “ G_{Φ_A} ”, “Fitzpatrick function”, “fixed point”, and “ q -positive” on arXiv, publisher pages, and author preprint pages. No prior statement of Theorem 1 was found. The source contains the metric criterion and the coordinate-axes example but does not state the conic generalization. Novelty confidence is moderate because the argument is short and close to that example.

Human-review recommendation. Verify the conjugate calculation, the use of $\lambda A = A$ for $\lambda > 0$, and the literature search before making a novelty claim.

References

- [1] Y. García Ramos, J. E. Martínez-Legaz, and S. Simons, *New results on q -positivity*, Positivity 16 (2012), 543–563, doi:10.1007/s11117-012-0191-7; arXiv:1111.6094.
- [2] S. Fitzpatrick, *Representing monotone operators by convex functions*, Workshop/Miniconference on Functional Analysis and Optimization, Proc. Centre Math. Anal. Austral. Nat. Univ. 20 (1988), 59–65.